COPYRIGHT RESERVED BCA (Sem-I) – Mathematics

(103) GE-I

2024

Time : $1\frac{1}{2}$ hours

Full Marks : 40

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer **all** sections as directed.*

Section–A

1. Answer any **four** questions of the following : 6×4=24

(a) State and prove Leibnitz theorem.

(b) State and prove Maclaurin theorem.

(c) Find the maximum and minimum value of $x^3 - 12x^2 + 45x$ in the interval [0,7].

(d) Explain the method of finding the solution of linear differential equation of first order.

(e) Explain the method of finding expression of Sin$x$ in Maclaurin's series.

(e) Prove that

$$\int_0^{2a} f(x)dx = \begin{cases} 2\int_0^a f(x)dx & \text{if } f(2a-x)=f(x) \\ 0 & \text{if } f(2a-x)=-f(x) \end{cases}$$

Section–B

2. Answer any **four** questions of the following : 4×4=16

(a) If $y = e^{a\sin^{-1}x}$, prove that

$$(1-x^2)y_2 - xy_1 - a^2y = 0$$

(b) If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, prove that

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sin 2u$$

(c) Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{a^2\sin^2 x + b^2\cos^2 x}$ or

$$\int_0^{\frac{\pi}{2}} \log \sin x dx$$

(d) Apply Maclaurin's theorem to prove that

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \ldots\ldots\infty$$

(e) Solve any **one** of the following :

(i) $\frac{dy}{dx} + 1 = e^{x-y}$

(ii) $\frac{d^2y}{dx^2} + b^2y = x\cos x$

(f) Solve any **one** of the following :

(i) $x^2\frac{d^2y}{dx^2} + 5x\frac{dy}{dx} + 4y = x^4$

(ii) $\frac{d^2y}{dx^2} + y = \sin 2x$

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